DemoCaP: One-Shot Learning from Demonstrations with Code-as-Policy Generation

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Abstract: Code-as-Policy (CaP) uses large language models to synthesize executable robot programs from language, but instructions often omit the geometric and procedural details needed for manipulation. We study whether one successful teleoperated demonstration can improve a language-only CaP agent without task-specific policy training. Our method, DemoCaP, identifies object interactions and extracts object-relative motion and spatial context from the demonstration, then uses it to improve code generation at test-time. We compare this method to CaP++, a language-only baseline we build based on CaP-X. Across the 12 conditions, DemoCaP achieves 57.8% mean success compared with 51.5% for CaP++. The gains persist under changes to object positions and task specifications, suggesting that a single demonstration can help close the specification gap while preserving the inspectability and editability of code-based policies. Further details and experiments are made available at egecaglar.github.io/democap-corl-late-breaking

Keywords: Learning from Demonstration, Code-as-Policy, Agentic Robotics

1 Introduction

Code-as-Policy (CaP) agents compose perception and manipulation tools into executable programs from language [1, 2]. While they have the advantage that their explicit control flow can be inspected and edited, their instruction inputs often omit grasp geometry, approach direction, and action ordering. We argue that a single demonstration can supply this missing information. While prior work generates programs from demonstrations [3, 4, 5], we ask whether one demonstration improves an otherwise unadapted, language-only CaP agent without task-specific training or repeated trial-and-error at test time.

DemoCaP augments our CaP-X-based language-only baseline, **CaP++**, with structured object interactions and object-relative trajectories extracted from one teleoperated demonstration. A VLM model then uses this context to revise its initial program and transfer motions to perturbed scenes. We compare DemoCaP with CaP++, CaP-X, VLA policies, and direct demonstration-frame conditioning.

2 Method

Setting and CaP++ baseline. We consider tabletop manipulation with a parallel gripper, fixed scene and wrist RGB-D cameras, and a language instruction. The agent generates a Python program from an instruction and initial image. CaP++ expands the CaP-X/CaP-Gym tool interface to 31 functions covering SAM 3 segmentation, VLM grounding, Contact-GraspNet grasp proposals, geometry, and high-level placement and articulation skills. Using the API functions, 13 high-level skills, focusing on tracking a target, planning for picks, places, and nonprehensile pull/push primitives, were derived

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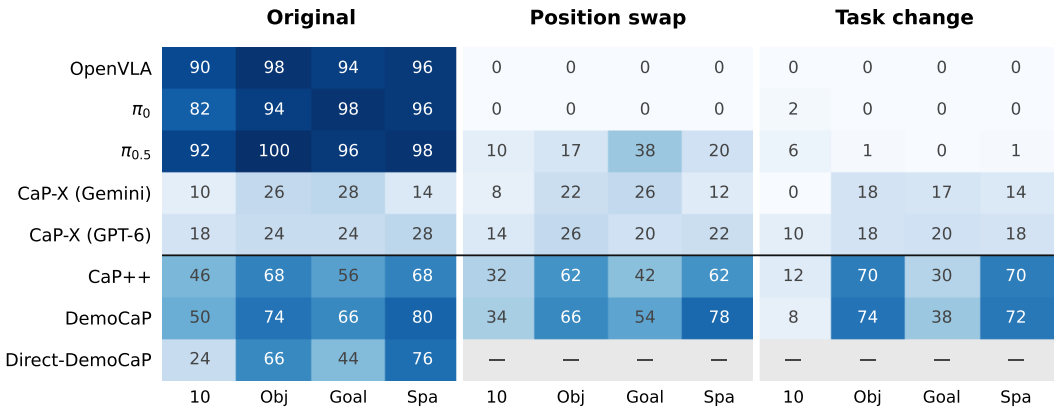


Figure 1: Success rates (%) across four LIBERO suites and three conditions (50 trials per task). Direct-DemoCaP was evaluated only on original tasks (gray cells: not evaluated). Some CaP-X/VLA results are taken from prior work. Obj=Object; Spa=Spatial.

with joint human and LLM feedback on 150 LIBERO-10 rollouts. The skill set was then fixed during evaluation, forming the CaP++ harness.

Demonstration parsing. Given a successful teleoperated RGB-D rollout, DemoCaP identifies manipulated and reference objects, and detects interaction intervals using gripper closure, end-effector proximity, and object motion. It extracts object-relative grasp/place/push trajectories, object bounding boxes, and spatial relationships to reference objects into a structured JSON representation.

Program conditioning. At test-time, Demo-CaP provides both the new scene's observations and the representation above to a VLM, alongside six new skills for analysis of the demonstration, which include skills for mapping relative approach positions to a new object instance, checking if a remapped path collides with an object in the new observed scene or resolving ambiguous objects by checking segmented images of objects from the demonstration. Thus, demonstration-conditioned code can transfer task structure to different configurations without blindly replaying world-frame trajectories. As an ablation, *Direct-DemoCaP* instead conditions the coding model simply on 16 sampled demonstration frames.

3 Experimental Results

We evaluate four LIBERO suites across original scenes, position swaps, and task changes (12 conditions; 50 trials per task), alongside LIBERO-PRO perturbations [6, 7]. One demonstration is provided per task, with new demonstrations for changed goals.

Across the 12 conditions (Fig. 1), DemoCaP achieves 57.8% mean success versus 51.5% for CaP++, improving 11 of 12 conditions. Both methods outperform the CaP-X baselines, while DemoCaP also exceeds Direct-DemoCaP on all four original suites. Table 1 shows that DemoCaP improves average performance across all evaluated perturbation categories, although not every individual suite benefits.

These results suggest that structured information from a single demonstration can improve code-based manipulation without task-specific policy training. We provide further results in the supplementary material; assessing real-world robustness remains future work.

	Original	Swap	Appearance	Distractor	Language	Environment	Shift†
CaP++	59.5	50.8	45.0	51.8	62.0	44.5	60.0
DemoCaP	**67.5**	**56.0**	**53.3**	**58.5**	**65.0**	**49.5**	**67.0**

Table 1: Success (%) averaged equally over four suites; †Object only.

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